Question 1: Differential Equations

Find the general solution to the second-order homogeneous differential equation $y'' + 4y = 0$.

Question 2: Linear Algebra

Determine the eigenvalues and corresponding eigenvectors for the matrix A = [[3, 1], [0, 2]].

Question 3: Multivariable Calculus

Evaluate the surface integral of vector field F over the closed unit sphere centered at origin.